Decision Trees — Lab Assignment

Goal. Implement Decision Trees from scratch and understand impurity measures (entropy, Gini), threshold selection for continuous features, overfitting control via depth and leaf-size. (No scikit-learn for the decision-tree training itself.)

Formulas

Entropy

$$H = -\sum_{i=1}^{C} p_i \log_2 p_i$$

where $p_i = \frac{\text{count of class } i}{\text{total samples in node}}$ and C is the number of classes.

Gini Index

$$G = 1 - \sum_{i=1}^{C} p_i^2$$

where p_i is defined as above.

Weighted Impurity for a Split

$$w\_imp = \frac{n_L}{n} \cdot imp_L + \frac{n_R}{n} \cdot imp_R$$

where n_L and n_R are the number of samples in the left and right child nodes, n is the total number of samples, and imp_L, imp_R are the impurity measures (entropy or Gini) for left and right nodes.

Information Gain / Reduction in Impurity

$$Gain = imp_{total} - w\_imp$$

where imp_{total} is the impurity of the parent node.

Threshold Selection for Continuous Feature

$$threshold = \frac{x_i + x_{i+1}}{2}$$

where x_i and x_{i+1} are consecutive sorted feature values in the node.

Exercise 1 — Decision Tree (from scratch) with Continuous Features

Task. Implement a binary decision tree learner:

(a) Splitting criteria: support both **entropy** and **Gini**.

(b) Continuous features: evaluate candidate thresholds at midpoints of sorted unique values; pick the split that maximizes impurity reduction.

(c) Stopping: stop if node is pure, or `max_depth` reached, or `min_samples_leaf` reached.

(d) Train two trees with the same settings (`max_depth`, `min_samples_leaf`), one with entropy and one with Gini.

Hyperparameter search (validation). Grid-search `max_depth` $\in \{2, 3, 4, 5, 6\}$ and `min_samples_leaf` $\in \{1, 3, 5, 10\}$ Pick the best per-criterion model based on validation accuracy, and report its test accuracy.

1 Sample Test Case

```
input = 10
1.382693181 0.14339787 1
-1.555508966 -0.095964842 0
1.316927777 0.934459346 1
1.168679898 -0.056381458 1
0.30005492 0.637478304 1
1.41811695 0.305671002 1
-0.122248166 0.805312495 1
1.760987628 2.009705229 1
-1.16710613 0.359100776 0
1.519941217 1.826766108 1
```

```
output =
Best (entropy): depth=2, minleaf=1, val_acc=0.77, test_acc=0.83
Best (gini):    depth=2, minleaf=1, val_acc=0.77, test_acc=0.83
```